Indian Statistical Institute, Bangalore

B. Math.

First Year, First Semester

Analysis I

Back paper examination — Date : Dec. 23, 2024

Total Marks: 105 — Time: 3 hours

Maximum marks: 100 — Instructor: B V Rajarama Bhat

(1) Let F be the collection of all finite subsets of the set of natural numbers. Is F countable or uncountable? Prove your claim. [15]

(2) Show that every sequence of real numbers has a monotonic subsequence. [15]

(3) Let a be a real number with $a > 1$. Show that

$$\lim_{n\to\infty} a^{\frac{1}{n}} = 1.$$

[15]

(4) Let A, B be non-empty subsets of $\mathbb{R}$ and let $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$ be functions. Suppose $f(A) \subseteq B$ and $h = g \circ f$. Fix $c \in A$. Suppose f is continuous at c and g is continuous at $f(c)$. Show that h is continuous at c. [15]

(5) Let d be a real number. Suppose $u : \mathbb{R} \to \mathbb{R}$ is a function differentiable at d. Define $v : \mathbb{R} \to \mathbb{R}$ by $v(x) = |u(x)|$, $\forall x \in \mathbb{R}$. If $u(d) \neq 0$, show that v is differentiable at d. [15]

(6) Show that if a series of real numbers $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then $\sum_{n=1}^{\infty} a_n^3$ is absolutely convergent.

(7) State and prove mean value theorem. [15]